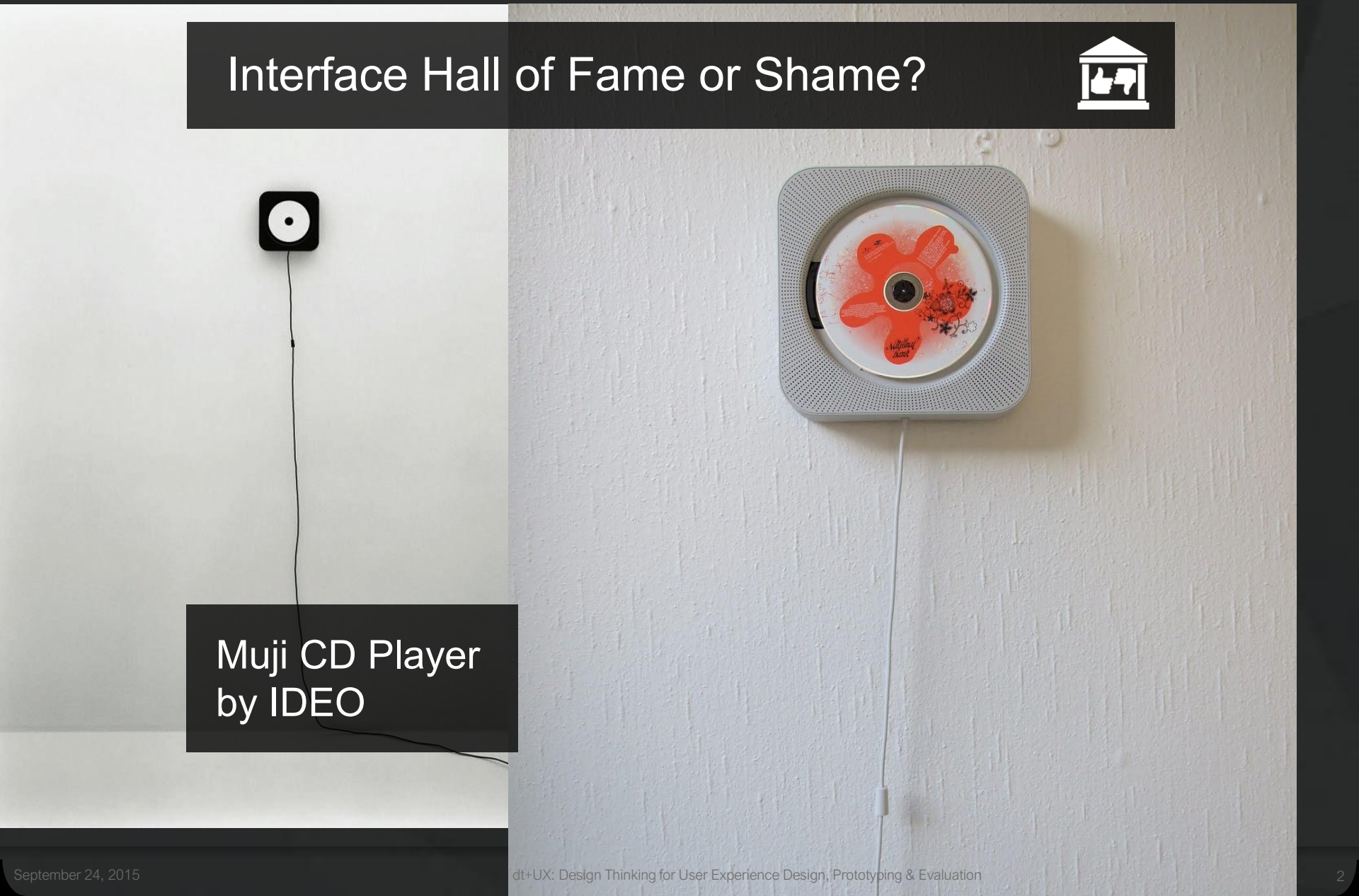
Human-Computer Interaction

4. Data Collection

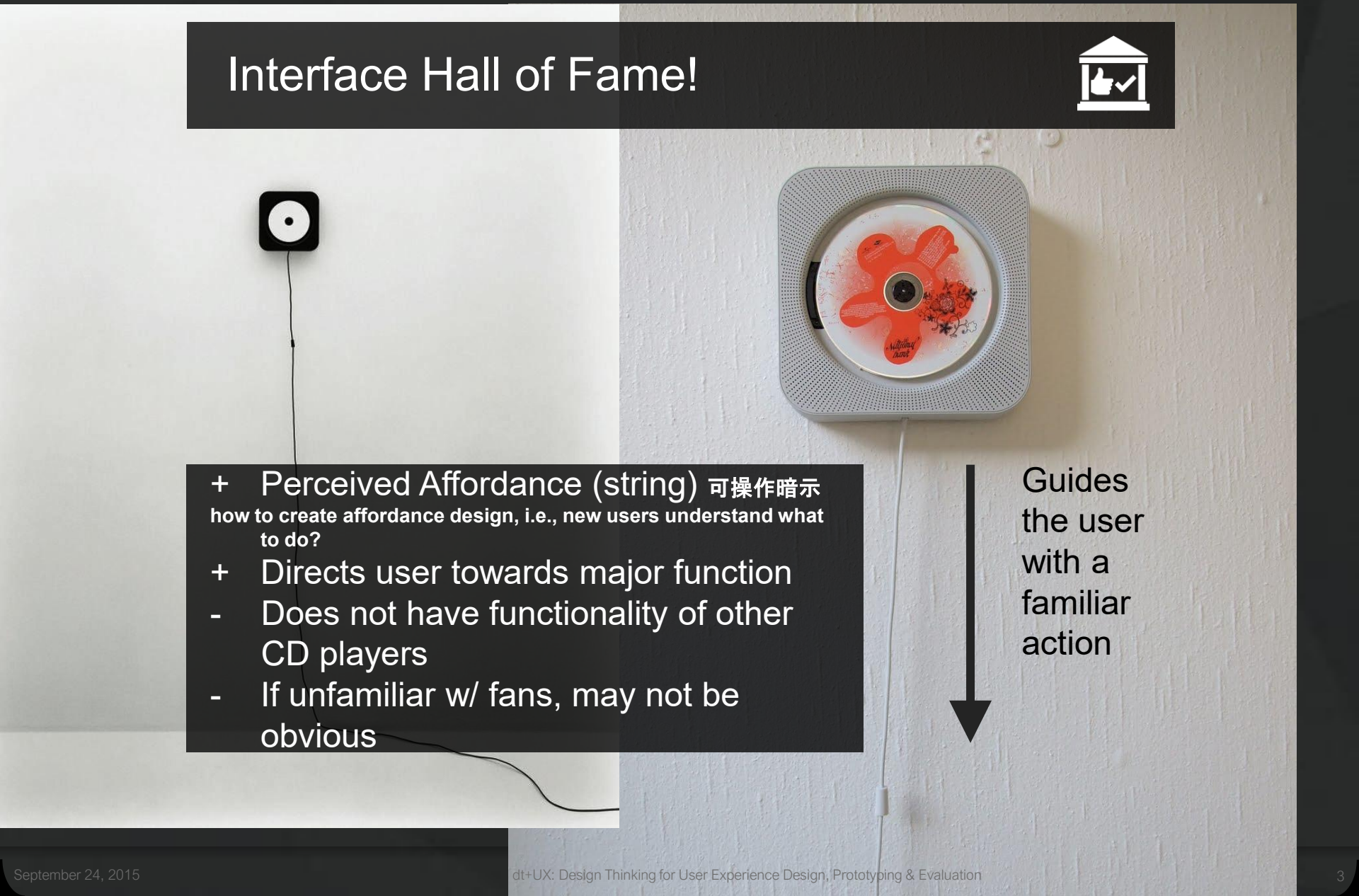


Interface Hall of Fame or Shame?



Muji CD Player
by IDEO

Interface Hall of Fame!



+ Perceived Affordance (string) 可操作暗示
  how to create affordance design, i.e., new users understand what to do?
+ Directs user towards major function
- Does not have functionality of other CD players
- If unfamiliar w/ fans, may not be obvious

Guides the user with a familiar action

Last week

- Understanding users: User analysis
- Understanding tasks: Task analysis

多选题 1分

设置

作为任务分析的可交付形式，下列哪些选项的说法是正确的?

A Table: User groups X tasks

B Table: Objects X tasks

C Sequence diagrams (XML)

D Task profiles

提交

主观题 10分

设置

为什么要关系task frequency?

作答

Recap: Task Analysis

- "Define" the needs
- The process of examining the way in which people perform their tasks

Task analysis is useful for understanding:

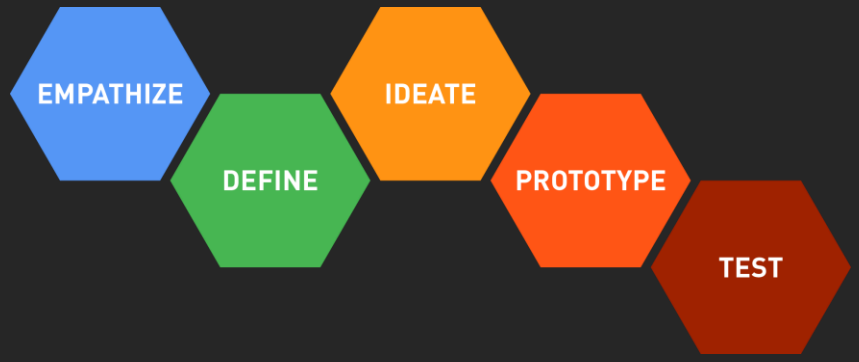
- Users' goals and what they are trying to achieve
- The steps that your users currently take in order to achieve their goals
- The personal, social and cultural experiences that users bring to the tasks
- The influence of the physical environment on the users while attempting to meet a goal

Recap: Task Analysis: Steps

1. Identify the task to be analyzed: What is that user's goal and motivation for achieving it?
2. Break this goal (high-level task) down into subtasks: You should have around 4–8 subtasks after this process. If you have more, then it means that your identified goal is too high-level and possibly too abstract
3. Collect data to identify tasks/subtasks that the user performs
4. Draw a layered task diagram of each subtask and ensure it is complete.

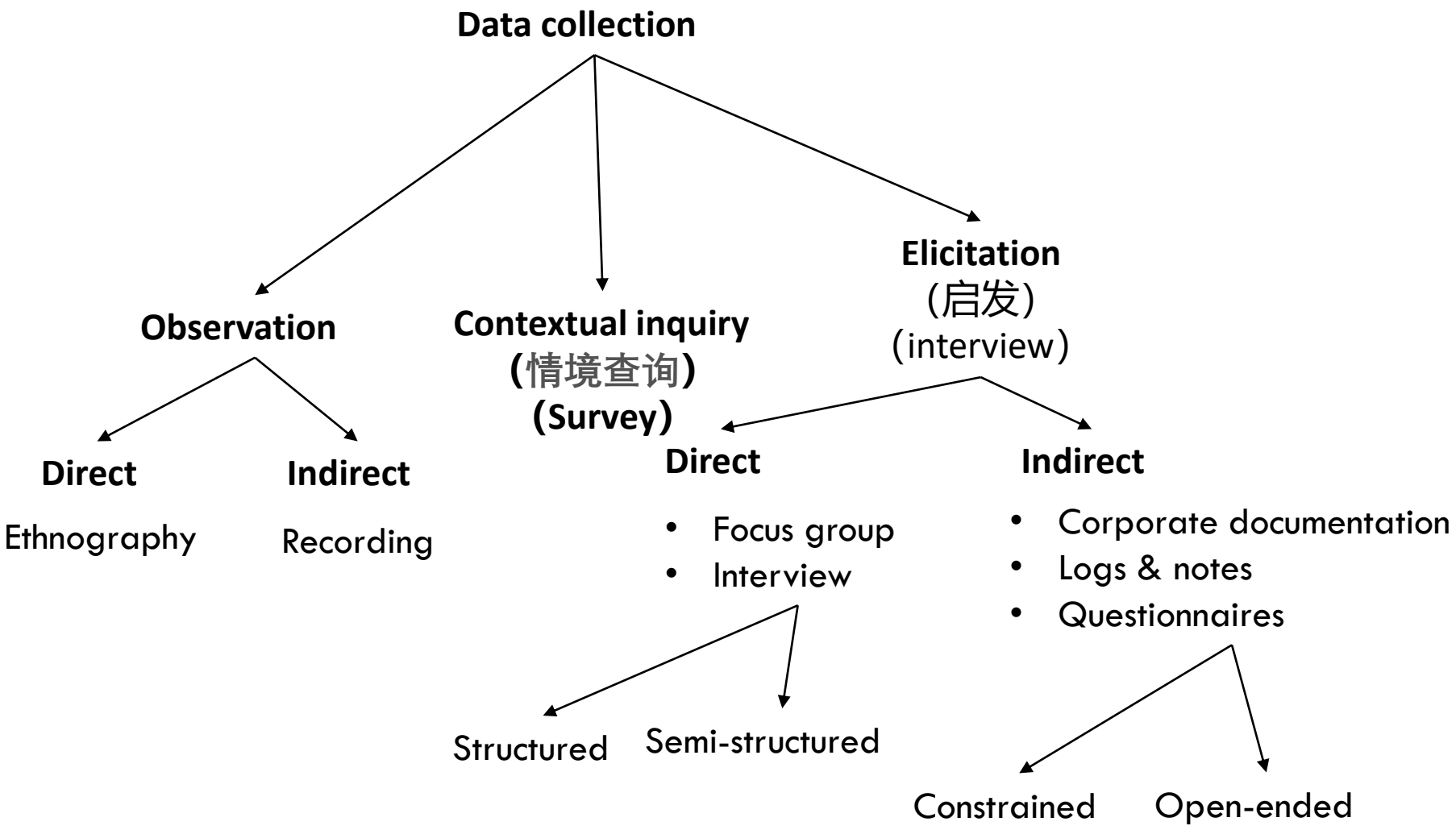
Today

- Data collection method
  1. Observation
  2. Interview
  3. Survey



Define:
Collecting Data from Users

Data collection methods



1. Observation



- Direct
  - Field studies: directly observing users in their usual work or home environment, doing normal work or tasks
  - Controlled studies: directly observing users in a controlled setting, performing special tasks
- Indirect
  - Video recording/performance logging
- Commonly used for task analysis or performance evaluation

2. Interview



A method of asking questions & listening

- Ask what you can't observe
- Hear about people's thoughts and experiences:
  - What they do
  - How they do things
  - Their opinions on current activities
  - How much they like one thing compared with another

- Interviewing skills are a basis for more advanced methods - survey design, contextual inquiry, task analysis, and so forth.

Types of Interviews

- Structured
  - Pre-determined questions to ask in a set way

- Semi-structured
  - Start with set topics for discussion and follow up based on the answers
  - Useful for gathering requirements and understanding users' opinions further in depth
  - One of the most widely used data collection methods in UI/UX design
  - From now on, I mean semi-structured interview when I say interview

How many users for Interview?

- As many as you can afford
- As many as you need before you stop finding new things out
- **10-15** is a good number

Make sure to choose representative users.

→ Who are the representative users?

Pros of Interview

- “Deep” rich data
- Interactive
- Open-ended
- Exploratory
- More economical in some cases than observing the user

Cons of Interview

- Time-consuming (both in conducting interviews and doing analyses)
- Can be inconsistent and subjective
- Amount of data can overwhelm
- Interviewee can be biased by interviewer, or what they say might be different from what they do

Interview Stages

1. **Plan the interview**
2. Conduct the interview

Planning an Interview

How much structure are you going to use?

- Structured
- Semi-structured

You will want to write down questions or notes to guide your thinking.

Questions to Ask

- What they do for a certain goal
- How they do things for the goal
- Their opinions about their current activities for the goal
- Their complaints about their current activities
- How much they are satisfied with one thing that they know compared with another thing that they know

But before all of these, you should have a clear definition of who “they” are!

Formulating Questions

1. Start by defining broader themes

- Think through what you are trying to get out of the interviews
- Think about themes you are trying to uncover, not specific questions just yet

Examples

- "Why do people shop online?"
- "How do people shop online?"
- "For your customers, what is the difference between online and offline shopping?"

Formulating Questions

2. Break down your questions to make them answerable

Example

“Why do people shop online?”

To:

- “What types of product do you buy online?”
- “What types of product do you avoid buying online? Why?”
- “What do you like the most and the least about the checkout process?”

Formulating Questions

3. Prioritize open-ended questions

Give users some room to elaborate their answers, as opposed to making super binary questions.

Example

- “What was the last thing you bought online?” vs.
- “Tell me about the last time you bought something online.”

Get specific

- Ask for tasks, roles and details of tasks
- Ask about specific moment in the past: walk through tasks “yesterday”
- Ask for examples!!!

Example.

- “What goes through your head when an online purchase fails?” vs.
- “Tell me what went through your head the last time you tried to buy something online and the purchase failed.”

Things People Know/Don't Know

People know

- What they do
- How they do things
- Their opinions about their current activities
- Their complaints about their current activities
- How much they like one thing that they know compared with another thing that they know

But, people don't know enough of

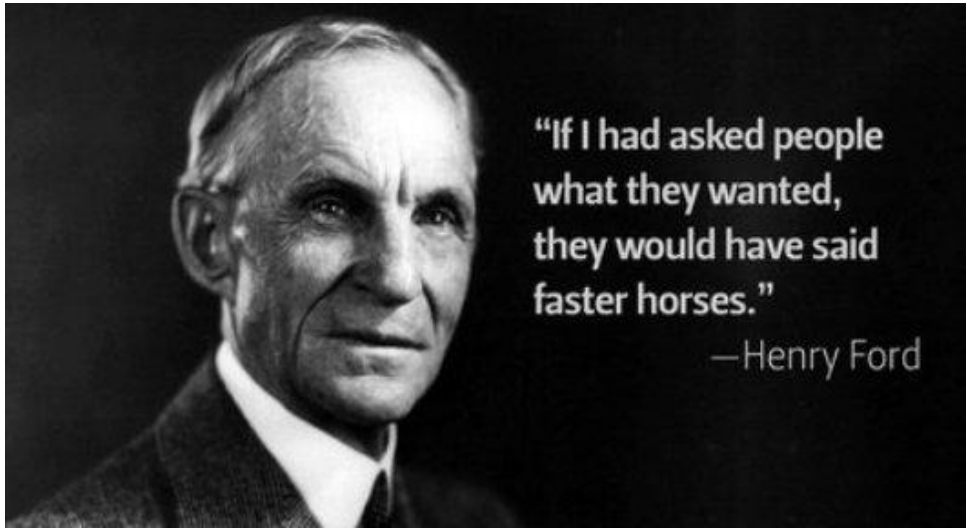
- What they would do/like/want
- How often they do things
- How much they like things on an absolute scale. If you ask, they will "guestimate"

Don’t ask:

- How much they like things on an absolute scale. If you ask, they will “guestimate”
- Questions that you can easily predict the answers
- Questions that will influence the answer
  - Don’t ask leading questions
  - Example:
    - “How anxious do you feel when an online purchase can’t be completed successfully?”
    - “Try to remember the last time an online purchase couldn’t be completed for some reason. How did you feel then?”

Don’t ask:

- How much they like things on an absolute scale. If you ask, they will “guestimate”
- Questions that you can easily predict the answers
- Questions that will influence the answer
- What they would do/like/want; new features



Be cautious about asking...

Anything that makes people imagine hypothetical situations

- What features people would like: they will focus on functionality they currently understand
- How to design a user interface: they do not know what is available

A：我唯一的烦恼是当服务器宕机时，一切都会冻结。

Q-：当我遇到这种情况时，我感到非常紧张。
Q+：那么你会怎么做呢？

A：是的，我在上面工作过的东西真的会在它重新启动的时候还在吗？

Q-：我知道，我希望在一切都冻结起来的时候能有一些指示，这样你就知道你的工作没有丢失。
Q+：当你遇到这种情况时，你的反应是什么？

提出无偏见的问题

-问：您经常使用哪些桌面应用程序？看起来像是文件管理器、日历管理器…

避免提供选项

答：我经常使用日历管理器和邮件工具。

问-：您是使用它来与您的工作伙伴安排约会，还是自己单独跟踪？

问+：您使用日历管理器来做什么？

与全职妈妈Amy的访谈

采访者：如果您在前方有一个麦克风，并且有一个小面板可以录制东西，您觉得这会有用还是没用？

Amy：我不确定我会用它来做什么……

采访者：哦，那您希望可以有什么东西呢？

Amy：我希望有一个我可以与之交谈并为我做事的东西（笑）

采访者：是的，那您需要……

Amy：拿开这个麦克风！

采访者：没错。

Create Interview Script

1. Know what you want answered: Write down a problem statement

   Examples of Problems Statements:

   - "Why do people make purchases online?"
   - "How do people make purchases online?"
   - "What's the difference between online and offline purchasing for the consumer?"

2. Reframe your problem statements: think of different perspectives to express the same problem statements.

   - Original: "Why do people make purchases online?"
   - Reframe 1: "What makes people want to buy online?"
   - Reframes 2 & 3: "What makes a product saleable online?", "What makes a product unsaleable online?"

Create Interview Script

3. Build a list of questions based on your reframed problem statements
   1) Don't ask leading questions.
      - Leading Question: "How angry do you usually feel when an online transaction fails to go through successfully?"
      - Non-leading Question: "Recall a time when an online transaction failed to go through successfully. How did you feel?"
      - Even More Non-leading Question: "Recall a time when an online transaction failed to go through successfully. What did you think or feel?"
   2) Ask about specific incidents in the past, when possible.
      - Speculative Question Based on Imagination: "Tell me what goes on in your head when an online transaction fails to go through successfully."
      - Context-based Question Grounded in Reality: "Tell me what went on in your head the last time an online transaction fails to go through successfully."
   3) Ask more open-ended questions.
      - Specific Question: "What was the last thing you bought online?"
      - Open-ended Question: "Tell me more about the last time you purchased something online."
   4) You can ask more than one question to find out the same thing.
      - Q1: "What are some reasons why you shop online?"
      - Q2: "What are some things that will make you decide not to buy from an online store?"

Create Interview Script

4. Paraphrase each question 2–3 times
5. Give the whole list of questions a structure
   1) Introduction

      Example: "Hi Alice, thanks for coming down. My name is _____, and I'll be conducting the interview today. We're doing some market research, so some of the questions might be a little personal, and will concern your lifestyle. If you feel uncomfortable answering any of the questions, just let me know and we can skip it. I can't tell you much about what we do or the exact reason we're doing this research now, because it might affect the way you answer the questions. But I can definitely tell you after the interview is over. Do you have any questions so far? If you don't mind, I would record the audio for this interview; it will only be used internally, for the purposes of this research."

   2) Warm up questions
      - Examples of warm up questions: "What's your occupation / what do you do all day?", "Can you tell me more about your hobbies?", "How often do you use the internet? etc."

   3) Main body questions
      - Start with questions relating to specific incidents in the past, before moving on to speculative ones.
      - Follow a flow that makes sense to the respondents

   4) Wrap up
      - Give your respondent a sense of closure.
      - Finally, thank them for their time and contribution towards your research.

Final Interview Script

Interview script

- Intro -

Hello user thank you for your time. I am collecting data for a research project pertaining student safety, is it alright if I use your response as part of my project? You may find some of my questions too personal, if you are uncomfortable with answering them just let me know and I will move on. Do you have any questions for me before I start?

- Warm Up -

How many classes are you taking this semester? How often do you commute from one campus to another?

Is this your first time on your own? If so how do you think you are handling it? If not what prior experiences have you had where you have been primarily responsible for your own well being?

- Main Body -

- Topic 1: Student Safety
  - Q1. Would you say that you felt prepared to go to college? Describe any new responsibilities or concerns that came as a surprise to you.

...

- Topic 3: Campus Safety and Concerns
  - Q1. Do you think that there are any specific areas (on or near campus) that could be unsafe for students?

    Follow Up: What is your overall opinion about the safety of this campus?
  - Q2. What do you think the campus security does to communicate with students in an emergency?
  - Q3. Does the campus security provide any way to connect them with students during an emergency? Do you think if it's effective?
  - Q4. Are there any safety measures that campus security takes to protect students who live on campus or off campus?
  - Q5. What do you think the campus is doing to spread safety awareness?

    Follow-up: What are your thoughts and concerns about the campus safety awareness?

- Closing -

How would you feel about a phone application that was able to alert your guardians / loved ones in the event of an emergency?

Follow Up: Do you think that you would personally use this kind of service? Why or why not?

Thank you for your responses!

Interview Stages

1. Plan the interview
2. **Conduct the interview**

主观题 10分

设置

课堂练习

两人一组，5分钟组织问题，3分钟interview+3分钟interview

主题：就业意向 （5-3-3）

作答

Group project

P2: Planning Data Collection

Design a semi-structure interview script and a list of survey questions that you will use to collect data from potential users.

How many questions? At least 3 topic areas (themes) and 5 questions per topic for an interview; at least 15 questions for a survey. But, you need to make sure that you have enough questions to know about the users, their context, and the task. If I find important topics (based on your proposal) are missing, you will loose points.

- Do not ask the same question in an interview and a survey
- You should think deeper why you are asking a question in an interview, not in a survey (and vice versa)

P2: Planning Data Collection

- Turn-in: an interview script & a list of survey questions
- Format:
  - One consolidated PDF file
  - On top: Put your team name and a brief description of your project
  - Style: 12 point scale in Times New Roman, 1.5 line spacing
  - At the bottom: State who contributed to the original writing of each section and who helped improve and/or review
- Due: Midnight 9/30

P2: Planning Data Collection

Rubrics

- Interview (2.5pt)
  - Structure: intro-main body-closing (1pt)
  - Quality of questions
    - Whether you ask relevant questions to your topic/problem (0.5 pt)
    - Whether you ask in-depth contextual questions (0.5pt)
    - Whether the answers to questions lead to in-depth, follow-up questions (0.5pt)
- Survey (2.5pt)
  - Whether you used proper types of questions (e.g., open ended vs. radio button) (1pt)
  - Whether you ask relevant questions to your topic/problem (1 pt)
  - Whether you ask appropriate demographic information questions (0.5pt)

P2: Planning Data Collection

Rubrics

- You will lose 50% if your submission does not follow the format
- You will lose 30% if it's a late submission (within 24 hours)
- You will lose 10% per missing an important topic
- You will lose 10% if the number of questions doesn't meet the minimum

Sample Interview Script

Interview script

- Intro -

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- Closing -

How would you feel about a phone application that was able to alert your guardians / loved ones in the event of an emergency?

Follow Up: Do you think that you would personally use this kind of service? Why or why not?

Thank you for your responses!

Sample Survey

Survey questions

- Q3. What is your marital status? (Radio buttons - Single, Married, Divorced, Other [with comments])
  - Single
  - Married
  - Divorced
  - Other (Explain) _____
- Q4. What is your employment status? (Radio buttons - Unemployed, Salaried Worker, Wage Worker, Self-Employed, Other [with comments])
  - Unemployed
  - Salaried Worker
  - Wage Worker
  - Self-Employed
  - Other (Explain) _____
- Q5. How frequently do you contact your doctor's office to schedule an appointment? (Likert Scale - Daily, Weekly, Monthly, Annually)

  Daily Weekly Monthly Annually
- Q6. How do you typically contact your doctor's office? (Radio Buttons - Email, Phone Call, Text Message, Proprietary Application [with comments])
  - Email
  - Phone Call
  - Text Message
  - Proprietary Application (Explain) _____
- Q7. How responsive is your doctor's office in responding to your messages/voicemails/etc.? (Likert Scale - Very Unresponsive, Unresponsive, Average Responsive, Responsive, Very Responsive)

Very Unresponsive Unresponsive Neither Unresponsive/Responsive Responsive Very Responsive

- Q8. Who do you most often speak to when making a doctor's appointment? (Checkboxes - Doctor, Nurse, Nurse's assistant, Receptionist, Other [with comments])
  - ❑ Doctor
  - ❑ Nurse
  - ❑ Nurse's Assistant
  - ❑ Receptionist
  - ❑ Other (Explain) ____
- Q9. How many different apps do you typically use on your phone? (Range - 1 _ _ _ _ _ 10+)

  1 2 3 4 5 6 7 8 9 10 10+
- Q10. How comfortable are you with using technology in your everyday life? (Likert Scale - Very Uncomfortable, Uncomfortable, Neither Uncomfortable nor Comfortable, Comfortable, Very Comfortable)

Very Uncomfortable Uncomfortable Neither Uncomfortable/Comfortable Comfortable Very Comfortable

- Q11. Do you use your phone to store sensitive information? (Passwords, bills, bank statements, etc) (Open-ended - Explanation of what kind of sensitive info, parameters for storing certain info)

  _____
  _____
  _____
- Q12. How often do you use video chat software? (Google Hangouts, Skype) (Likert Scale - Daily, Weekly, Monthly, Annually, Sparsely)

  Daily Weekly Monthly Annually Sparsely
- Q13. Do you use your phone to track your health? (Fitness monitoring, diet tracking, etc) (Multiple Choice/Open-ended - If yes, Explanation of what kind of data is tracked, how it is tracked, etc.)